

HYBRID PV And THERMOELECTRIC DEVICE To Harvest SOLAR ENERGY More Efficiently

Hybrid power devices are combinations of different power technologies. Hybrid power plants often contain a renewable energy component such as photovoltaic (PV) that is combined with wind power, thermoelectric power, solar thermal power, or a system like battery storage or solar thermal storage. Thermoelectric generators are semiconductor devices that have no moving parts and convert heat directly into electricity. When combined with thermal storage they can provide electricity round the clock at as low as \$0.06 per kilowatt-hour and could achieve 16% efficiency.

PV cells convert the UV and visible regions of the solar spectrum while the thermoelectric modules use the infrared region to produce electrical energy. Thus, combining both these systems in a hybrid system provides enhanced performance. While PV panels convert up to 20% of solar energy into electricity, the solar thermal collectors capitalise on the untapped heat energy of the PV system, thereby increasing the energy production efficiency while occupying less space



RATHINDRA NATH BISWAS is a 1964-batch chemical engineering graduate from Jadavpur University, Kolkata. He was awarded a certificate for designing Benzol Plant by Giprokoks, USSR, and Certificate of Honour by Indian Institute of Metals. He has published 35 research papers in various journals. He retired from service as head - MECON, Durgapur.

Global energy demand is likely to increase by 48% in the next 20 years due to population explosion. Currently 80% of energy needs are met by fossil fuels, which emit greenhouse gases that lead to global warming and climate change. Their negative environmental impact is leading to development of renewable energy sources like solar, geothermal, and hydro.

Sun is an eternal source of renewable energy outputting 3.8×10^{20} megawatts (MW) of which our Earth receives about 1.7×10^{14} kilowatts (kW)—only a tiny fraction of the total radiation. However,

84 minutes of this tiny solar radiation is enough to meet our world's energy demand for one year (about 900EJ). India receives solar energy of 5 to 7 kilowatt-hours per metre square (kWh/m^2) for 300 to 330 days in a year, which is sufficient to install 20MW solar power plants per square kilometre land mass.

Solar energy can be utilised to heat fluids or to generate electricity. However, at present only 2% of world's energy comes from Sun, as solar energy harvesting has been expensive and its efficiency is low. New technological advances like improved solar cells, hybrid photovoltaic

and thermoelectric generators, use of aerogel, and cogeneration of electricity from solar cells as well as solar heating for thermal power generation is leading to decreasing cost and increasing efficiency,

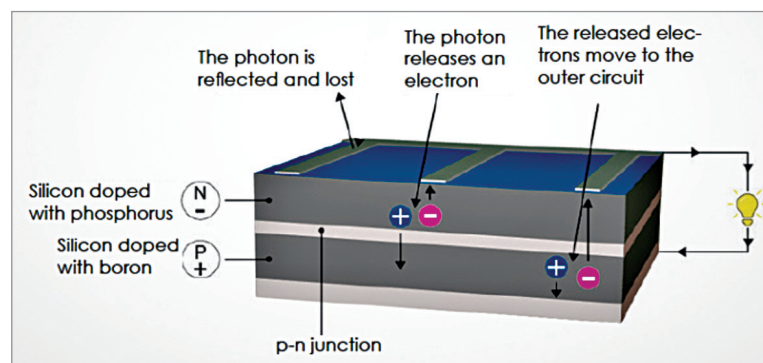


Fig. 1: Working of a PV cell